

Network Science

Extra: Dataset Cards

Luc Berthouze

L.Berthouze@sussex.ac.uk

University of Sussex

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Extra: Dataset Cards

- What is a Dataset Card
- Why are dataset cards important?
- Who is pushing this forward?
- How does this relate to Network Science?
- Network dataset card template

Dataset Cards

The idea of Dataset Cards is inspired by **model cards** (proposed by Google researchers for AI models, see [1]), but applied to datasets.

Dataset Cards are structured documentation meant to **improve transparency, reproducibility, and responsible usage** of datasets.

Open Images Extended - (MIAP)

Dataset Snapshot

PRIMARY DATA TYPE(S)

Non-Sensitive Public Data about people

DATASET SNAPSHOT

Total Instances	100,000
Training	70,000
Validation	7,410
Testing	22,590
Total boxes	454,331
Total labels	908,662
Average labels per image	9.08
Human annotated labels	All

DESCRIPTION OF CONTENT

Bounding boxes of people with perceived gender presentation attributes (predominantly feminine, predominantly masculine, unknown) and age range presentation attributes (young, middle, older, unknown). This adds nearly 100,000 new boxes that were not annotated under the original labeling pipeline of the core Open Images Dataset.

Note: All annotated images included at least one person bounding box in Open Images v6. 30,474 of the 100k images contain a MIAP-annotated bounding box with no corresponding annotation in Open Images. Almost 100,000 of the bounding boxes have no corresponding annotation in Open Images. Attributes were annotated for all boxes.

PRIMARY DATA MODALITY

Labels or Annotations

KNOWN CORRELATION(S)

- Gender presentation numbers are skewed towards predominantly perceived as **masculine & unknown**
- Age range presentation range numbers are skewed towards **middle**
- Perceived gender presentation is **unknown** for all bounding boxes with age range attribute annotated **young**

HOW TO INTERPRET A DATAPoint

Each datapoint includes a bounding box denoted by XMin, XMax, YMin, and YMax in normalized image coordinates. The next five attributes (IsOccluded through IsInsideOf) follow the [definitions from Open Images V6](#).

The last two values for each datapoint correspond to the gender presentation attribute and an age range presentation attribute, respectively.

Each annotation is linked to an Open Images key pointing to an image that can be found in [Common Visual Data Foundation \(CVDF\) repository](#).

Datapoint Example

EXAMPLE OF ACTUAL DATA POINT WITH DESCRIPTIONS

Field	Value	Description
ImageID	164b0e6d1fcf8e01	The image this box lives in
LabelName	/m/01q317	Labels are identified by MIDs (Machine-generated IDs) as can be found in Freebase or Google Knowledge Graph API . Label descriptions here
Confidence	1	A dummy value, always 1
XMin	0.897112	Normalized image coordinates indicating the leftmost pixel of the annotation
XMax	0.987385	Normalized image coordinates indicating the rightmost pixel of the annotation
YMin	0.615583	Normalized image coordinates indicating the topmost pixel of the annotation
YMax	0.895387	Normalized image coordinates indicating the bottommost pixel of the annotation
IsOccluded	0	Binary value indicating if the object is occluded by another object in the image
IsTruncated	1	Binary value indicating if the object extends beyond the boundary of the image
IsGroupOf	0	Binary value indicating if the box spans a group of objects
IsDepictionOf	1	Binary value indicating if the object is a depiction and not a real physical instance
IsInsideOf	1	Binary value indicating if the image is taken from the inside of the object
IsInsideOf	1	Binary value indicating if the image is taken from the inside of the object
GenderPresentation	Predominantly Masculine	Indicates the perceived gender presentation of the subject assessed by a third party
AgePresentation	Middle	Indicates the perceived age range of the subject assessed by a third party

Data Collection

DATACOLLECTION METHOD(S)

Derived

Vendor Collection Efforts

DATASOURCES BY COLLECTION METHOD(S)

Images	Open Images V6
Labels	Human annotators
Bounding Boxes	Human annotators

SUMMARIES OF DATACOLLECTION METHODS

100,000 images randomly sampled from the positive set of Open Images V6, which contains approximately 9.9M images

- Training Set: 70,000 sampled from 9,011,219 images
- Testing/Validation: 30,000 sampled from 167,056 images

EXCLUDED DATA

No excluded data

DATASELECTION CRITERIA - SCRAPING

- Images were sampled from the positive subset of training and testing/validation containing annotator-verified image labels
- Images contained at least one of five person classes (**man**, **woman**, **boy**, **girl**, or **person**)

Note: We did not include non-binary as a class label as it is not possible to label gender identity from images. Gender identity should only be used in situations where participants are able to self-report gender.

What is a Dataset Card?

It is a structured summary that describes key characteristics of a dataset. Its purpose is to provide a clear and concise overview, making it easier for researchers and practitioners to understand the dataset's context and applicability.

- **Basic information:** Name, source, and creators.
- **Motivation:** Why the dataset was created and what it is intended for.
- **Composition:** What kinds of data it includes (e.g., nodes and edges for networks, text, images, etc.).
- **Collection process:** How the data was collected, whether it is synthetic or real-world, and any preprocessing steps.
- **Uses:** Recommended applications and potential misuse cases.
- **Ethical considerations:** Biases, limitations, and any legal/privacy concerns.
- **Statistics and structure:** Key metadata like number of samples, distributions, missing values, etc.

Why are dataset cards important?

Dataset cards play a crucial role in promoting several key aspects of research:

- **Transparency:** Users can better understand what the dataset contains, fostering *trust*
- **Reproducibility:** By documenting the collection process and methodology, dataset cards enable others to replicate studies more easily.
- **Fairness & Ethics:** Highlighting biases and limitations encourages responsible data use (and minimise biases).
- **Better Decision-Making:** Helps researchers make informed decisions about dataset applicability.

Who is pushing this forward?

Several key organizations and initiatives are advancing the use of dataset cards:

- **Google Research** and **MIT-IBM Watson AI Lab** introduced the idea formally
- **Hugging Face** has implemented Dataset Cards in their dataset repository
- **OpenML** and **FAIR (Findable, Accessible, Interoperable, Reusable) Data** initiatives also promote structured dataset documentation, advocating for transparency and reproducibility in research

Network Dataset Cards: Specific considerations

When adapting dataset cards for network datasets, several specific aspects should be included:

- **Network type:** Social, biological, transportation, etc.
- **Nodes and edges:** What entities they represent
- **Degree distribution:** Whether it follows a power law or another characteristic distribution
- **Density and connectivity:** Global clustering coefficient, diameter, etc.
- **Data collection method:** Crawled, simulated, or manually curated
- **Ethical concerns:** If the data involves sensitive human interactions

Note

For more on this topic, consider reading [\[2\]](#) and [\[3\]](#)

A possible template (1/4)

1. Dataset overview

- **Dataset Name:**
- **Short description:** *(Briefly describe what the dataset represents, e.g., a social network, citation graph, or biological network.)*
- **Source:** *(Where was this dataset obtained? Provide URLs or references if applicable.)*
- **Creators / Curators:** *(Who collected or maintains this dataset? If known.)*
- **License:** *(Is it open-source? Any usage restrictions?)*
- **Date of collection / Last update:**

2. Motivation & Purpose

- **Primary purpose:** *(Why was this dataset created? What kind of analysis is it useful for?)*
- **Intended use cases:** *(List some applications, e.g., community detection, epidemic modeling, recommendation systems.)*
- **Known misuse cases:** *(How could this dataset be misused? E.g., privacy risks, unethical inferences.)*

A possible template (2/4)

3. Dataset composition

- **Network type:** (*Directed/Undirected, Weighted/Unweighted, Multiplex, Bipartite, Temporal, etc.*)
- **Number of nodes:**
- **Number of edges:**
- **Edge density:** (*Ratio of existing edges to all possible edges.*)
- **Degree distribution:** (*Power-law, Poisson, etc. Include a plot if possible.*)
- **Clustering coefficient (global & local):**
- **Largest connected component size:**
- **Other key structural properties:** (*E.g., modularity, assortativity, diameter.*)

A possible template (3/4)

4. Data collection & Preprocessing

- **Collection method:** *(Crawling, surveys, logs, simulations, etc.)*
- **Preprocessing steps:** *(Any data cleaning, filtering, or anonymization applied?)*
- **Missing data handling:** *(How were missing nodes/edges handled?)*
- **Data quality concerns:** *(E.g., biases, noise, representativeness issues.)*

5. Ethical & Privacy Considerations

- **Does this dataset contain personally identifiable information (PII)?** *(Yes/No. If yes, how is it handled?)*
- **Potential biases:** *(Are some groups underrepresented or overrepresented?)*
- **Fairness considerations:** *(Could analyses based on this network lead to unfair or discriminatory outcomes?)*
- **Risk of harm:** *(Any risks in making this dataset public?)*

A possible template (4/4)

6. Suggested analysis & Benchmarking

- **Recommended benchmarks:** *(Any known results or standard tasks for this dataset?)*
- **Baseline statistics:** *(E.g., centrality measures, robustness metrics.)*
- **Known issues / limitations:** *(What should users be cautious about when interpreting results?)*

7. Citation & References

- **How to cite this dataset:** *(Provide citation in BibTeX, APA, or other standard format.)*
- **Related papers / Studies using this dataset:** *(If any.)*

8. Additional Notes

- *(Any other relevant information?)*

References

1. T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. D. III, & K. Crawford, Datasheets for datasets. (2021). <https://doi.org/10.48550/arXiv.1803.09010>.
2. J. Bagrow & Y.-Y. Ahn, Network cards: Concise, readable summaries of network data. *Applied Network Science*, **7** (2022) 84. <https://doi.org/10.1007/s41109-022-00514-7>.
3. X. Zheng, R. A. Rossi, N. Ahmed, & D. Moritz, Network report: A structured description for network datasets. (2022). <https://doi.org/10.48550/arXiv.2206.03635>.